

07 August 2026

FX Outlook

“Is this an intervention?”

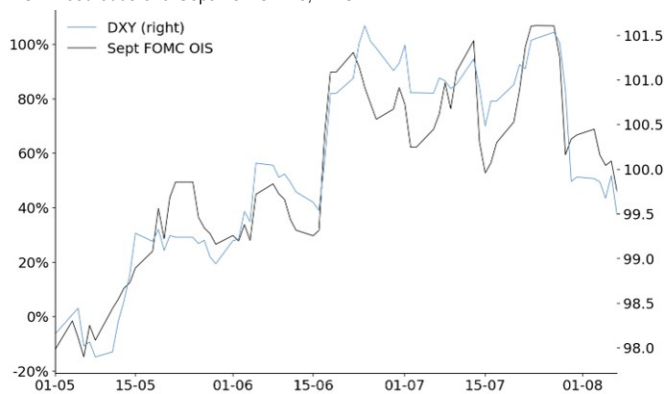
- We had been long USD since May, but neutralize our recommendation after payrolls. We see narrower ranges for USD based on recent data-flow.
- Data dependence still matters; the dollar continues to closely track near-term Fed expectations.
- Carry has proven resilient to several headwinds, and after neutralizing USD length, remains the central component of our trading strategy. Resilient growth, a pro-risk backdrop and still-high rate differentials argue against a full replay of August 2024.
- Large MoF intervention raises risks of alternatives to faster BoJ hikes. FIMA & GPIF have all moved into focus, following US Treasury intervention. We retain unchanged 164 YE targets for USD/JPY after intervention.
- We compare the 2024 & 2026 macro backdrops. Several ingredients behind the 2024 carry unwind remain, including USD downside risk, elevated US yields, and crowded positioning, but no BoJ surprise hike, better growth and lower political vol should limit the degree of high-vol deleveraging.
- Next week: RBA, Norges Bank, Riksbank; US CPI; Europe PMIs

Neutralizing USD length

We neutralize our long-USD recommendation (vs G10 low-yielders) in the wake of this week’s US nonfarm payrolls print. We turned long dollars in May on improved US data and participated through the ~100bp rally of real yields as Fed expectations were repriced. Over the last month, however, the dollar has failed to break the range while a mix of factors softened our confidence in the trade: 1) softer data (two consecutive NFPs, inflation data), 2) the latest détente in the US-Iran conflict (ToT), 3) better signs of global growth generally, 4) some evidence of renewed negative sensitivity to the UST long-end (Figure 2, Figure 3) and finally 5) last week’s joint US-Japan FX intervention. While we recognize risks around US CPI next week and we retain a mild dollar-bullish bias, we think **it’s prudent to neutralize our long-USD stance (vs G10 low-yielders) and instead concentrate risk in the long-carry trade**, which has so far withstood the joint intervention and yen appreciation.

Figure 1: USD continues to track Fed expectations, emphasizing data dependence

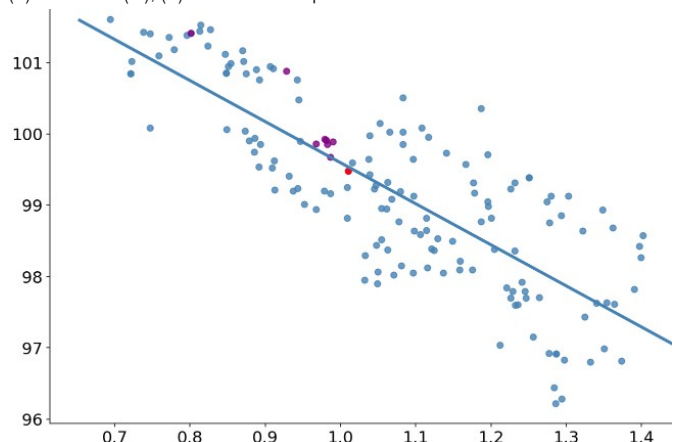
LHS: Priced odds of a Sept FOMC hike; RHS: DXY



Source: J.P. Morgan

Figure 2: USD also remains inversely correlated to curve steepness...

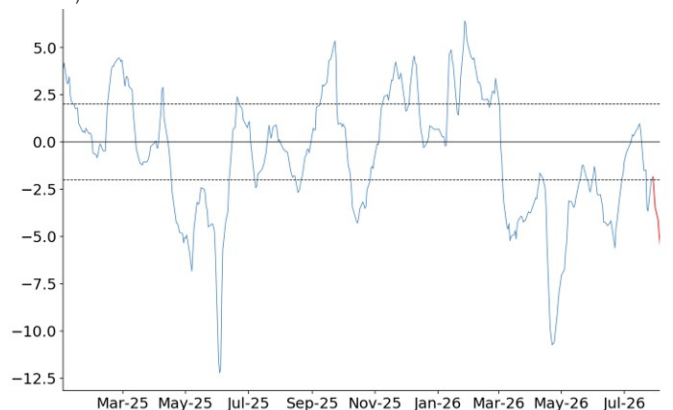
(X): US 2s30s (%); (Y): DXY. YTD. Purple since FOMC



Source: J.P. Morgan

Figure 3: ...reflecting renewed sensitivity to the long-end

1m rolling t-stat: USD sensitivity to 5s30s curve shape (controlling for 2Y US-G10 rate diffs). Red since FOMC



Source: J.P. Morgan

But data dependence will remain paramount, as Fed expectations are proving to be a solid anchor for USD in the short-term. The lack of forward guidance and some residual concern about the level of inflation should keep rates pricing on its toes throughout 2H. In that respect, it's important to note that DXY has closely tracked pricing of near-term Fed hiking expectations (Figure 3), which for now retains a decent premium still for a hike even after NFP. While we still think it's appropriate to unwind our dollar long recommendation as ranges are likely narrower even with CPI upcoming, the sensitivity of the dollar to Sept FOMC pricing underscores the renewed empirical importance of data dependence.

The latest on JPY, intervention and impacts (or lack thereof) on FX carry

The JPY intervention saga continues to evolve given several wrinkles this time compared to past episodes. Questions continue to revolve around - why UST intervened via EUR instead of USD, how much capacity does MoF have after such large-scale intervention this episode, and what other mechanisms can be deployed to support JPY in the absence of a more obvious hawkish pivot from BoJ. We've covered material ground in other research pieces this week, and synthesize the key takeaways across our geographic teams here:

- Coordinated intervention marks the first US participation in yen-buying operations since 1998 and suggests a **stronger-than-expected willingness by Washington** to support efforts to curb excessive yen weakness. The apparent use of EUR/JPY rather than USD/JPY reinforces the view that the objective was to support Japan's efforts to prevent yen depreciation, not to weaken the dollar. More broadly, the episode underscores that Japanese **authorities view USD/JPY above 160 as excessive, while remaining reluctant to rely on accelerated BoJ rate hikes** as the primary response. We **maintain our 164 year-end target**. See [Unexpectedly strong US-Japan cooperation against weak JPY](#) (2 August).
- The Fed's FIMA repo facility** can augment Japan's intervention toolkit, but was not designed for FX intervention and **cannot replace traditional intervention resources**. Unlike ESF or MoF assets, FIMA provides temporary USD financing rather than permanent intervention funding, limiting its effectiveness as a long-term solution. Expanding the facility outside periods of funding stress could also raise **broadier policy questions**. See [FIMA & FX intervention](#) (4 August).
- Japan's cumulative intervention this year has already far exceeded 2024's total, suggesting **traditional intervention tools are becoming increasingly constrained** (Figure 4). Yet the distance to outright Treasury sales or meaningful reliance on FIMA may still be longer than

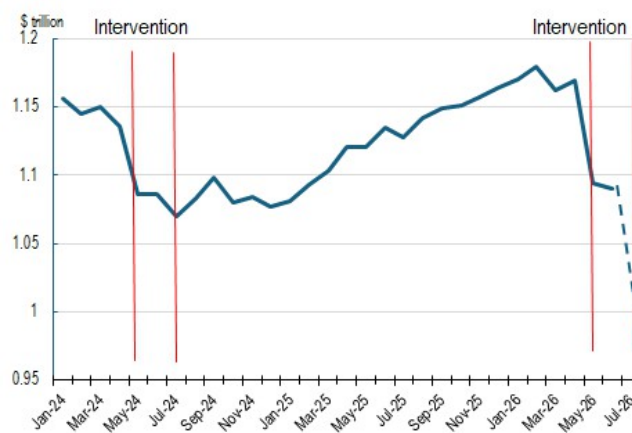
generally assumed, given the MoF's ability to generate cash from maturing reserve assets. Taken together, growing interest in coordinated intervention, FIMA, and GPIF allocation shifts suggests **authorities are searching for alternatives to an accelerated BoJ hiking cycle** (Figure 5) (we've added a hike in our BoJ '27 [call](#)), even if none appears likely to materially alter JPY supply-demand dynamics in the near term. See [Next move to stop JPY weakness: Not enough for a sash, much for a belt](#) (6 August).

- Despite a sharp oil selloff, equity volatility, and the first coordinated yen-buying intervention since 1998, **carry drawdowns remained modest and largely confined to JPY crosses, with little evidence of broader contagion** across the FX cross-section. Unlike August 2024, when carry deleveraging became systemic, high yielders generally held up well and low-yielder repricing remained contained, reinforcing our **constructive view on global carry baskets despite a more volatile macro backdrop**. See [FX Macro Quant: Carry resilient; maintaining a positive bias](#) (5 August).

The net result for our views is that 1) while joint intervention may temper upside risks to our USD/JPY 164 forecasts, it nevertheless does not fundamentally alter the yen-bear case, and we therefore keep our 164 target; and 2) we retain confidence in the global carry trade despite the vol in a key funder.

Figure 4: Developments in the outstanding amount of "Securities + Deposits" of Japan's FX reserves

Figure for July 2026 is JPM estimation.



Source: Japan Ministry of Finance, J.P. Morgan

07 August 2026

Figure 5: Comparison of tools to curb yen weakness

	MOF intervention	US joint intervention	GPIF	FIMA repo
Immediate availability	○	△	○	?
No need to sell US outright?	△	—	△	○
Potential ammunition	relatively small	relatively small in EUR/JPY	Up to \$190-222bn	\$60bn/day
memo		If they decide to move ahead with USD-selling, intervention capacity could increase substantially.	If it involves changing the basic portfolio, intervention capacity could increase. Compared with FX intervention, the flows would take longer to materialize, and the impact would be more dispersed over time.	Ultimately, the USD would need to be repaid. Cost is high

Source: J.P. Morgan, NY Fed

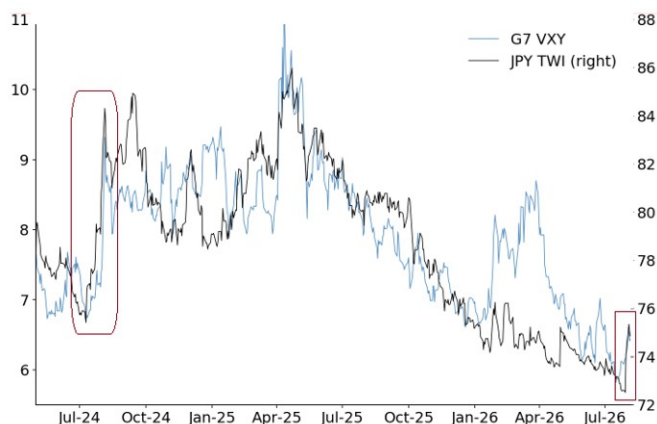
Lessons, similarities and differences vs 2024

Markets have taken recent yen appreciation in their stride, despite the largest intervention yet to date and with bad memories of the fallout from the summer of 2024.

With carry en vogue in 2026 and yen intervention coming in surprisingly strong, one might have reasonably expected more of a vol event in FX, sponsored by a deleveraging of the carry trade. Per Figure 6, that has not been the case. Indeed, as noted above and discussed further in this week's *FX Macro Quant*, that carry drawdowns have proven quite modest despite a toxic mix of factors bridging JPY intervention, tech equity drawdowns and oil price softness (see A. Delair, [Carry resilient, maintaining a positive bias](#)), while the JPY rally has been relatively modest and has not sponsored the same degree of vol spike in the broader FX space as we saw two years ago. For a more tactical quantitative comparison of backdrops around first slugs of JPY intervention, see Figure 7.

Figure 6: Simply in terms of magnitude, the JPY TWI rally has been more modest this time despite greater intervention volume, with limited spillover into broader FX vol

LHS: G7 VXY; RHS JPY TWI



Source: J.P. Morgan

Figure 7: Comparison of cross-asset performance around JPY 2024/2026 interventions. JPY appreciation is so far tracking that of 2024, but with less negative spillover elsewhere

Changes around July 2024 and July 2026 MoF JPY intervention episodes. 3m changes pre-intervention, and 1wk changes post-intervention. Pct chg or diff noted in index. Z-scores shown in parenthesis

	-3m		+1wk	
	2024	2026	2024	2026
Global FRI (Δ)	0.1 (0.3)	0.1 (0.3)	-0.1 (-1.1)	0.1 (0.5)
US FRI (Δ)	-0.1 (-0.2)	-0.0 (-0.2)	0.0 (-0.1)	--
US 2Y (Δ)	-0.35 (-1.0)	0.33 (0.2)	-0.18 (-1.5)	-0.09 (-0.8)
SPX (%)	9.2 (0.9)	2.5 (-0.1)	-0.8 (-0.5)	5.6 (2.4)
MSCI W (%)	6.9 (0.7)	3.0 (0.0)	-0.2 (-0.2)	5.0 (2.3)
BRENT (%)	-6.0 (-0.5)	-23.1 (-1.4)	0.0 (-0.1)	-12.4 (-2.5)
USD TWI (%)	1.4 (0.4)	1.1 (0.3)	-0.6 (-0.9)	-1.1 (-1.7)
DXY (%)	-0.2 (-0.2)	1.9 (0.4)	-1.2 (-1.4)	-1.2 (-1.3)
JPY (%)	-5.3 (-0.7)	-1.8 (-0.0)	3.5 (2.8)	3.6 (2.9)
FX VOL (Δ)	-0.02 (0.0)	-0.54 (-0.4)	0.25 (0.8)	0.21 (0.5)
USD POSITIONING (Δ)	29100 (0.3)	108159 (1.5)	-21126 (-1.4)	--

Source: J.P. Morgan, Bloomberg Finance L.P.

Below, we summarize the similarities and differences in the macro backdrop between 2024 and 2026. **Most actionable insights include: many of the preconditions for the 2024 carry unwind remain in place, particularly a USD downside shock, elevated US yields and crowded positioning.** That said, **the absence of a surprise BoJ tightening shock, a firmer global growth backdrop and less politically-induced volatility suggest the spillovers into broader carry and cross-asset markets should be materially less severe than in August 2024, even if further JPY appreciation remains possible.**

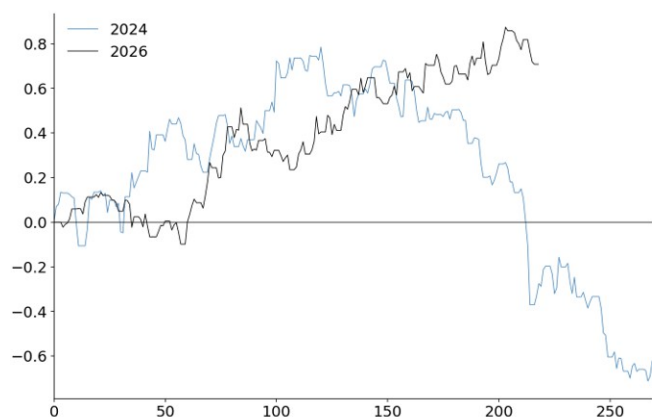
Similarities vs 2024

- **2024 intervention was relatively successful (USD/JPY to 140), and follows a similar playbook of intervening after a significant USD sell-off.** MoF intervention in 2024 is remembered for the tactical selling of USD/JPY on the back of a downside surprise in US CPI (and before the Fed started its easing cycle). While 2026 didn't necessitate quite the same immediacy from the MoF as the CPI date in 2024, they nevertheless intervened just one day after another significant dollar-down event (FOMC), which was a -1.3-sigma loss for DXY, with the USD losing more ground after NFP this week. **In a vacuum, this would suggest some caution in presuming JPY appreciation is over; JPY didn't peak until almost a full month after the initial intervention in 2024.**
- **There was a similar US yield uptrend between 2024 and 2026, both pressuring the yen and supporting the carry trade.** Following the hawkish rethink around the Warsh Fed throughout 2Q, US 2Y yields rallied about 80 points from the lows. 2024 featured a similar such rally, also approximating 80bps in the 2Y sector by mid-year, reflecting another episode of US exceptionalism and

largely unwinding the collapse in yields seen in 4Q'23 (Figure 9). **To the extent carry benefits from higher rates / less Fed easing, this seems material; we remain constructive on carry and US growth remains solid, but a collapse in yields (not the base-case) could exacerbate that trade and force further yen deleveraging.**

Figure 8: US yields actually rose by a similar amount into 2024

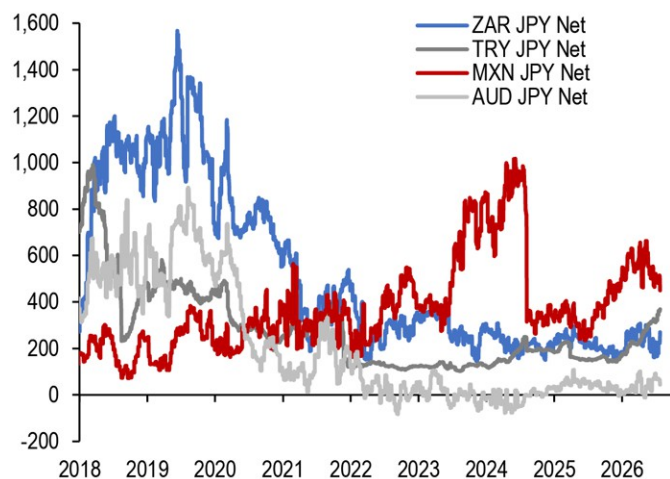
Cumulative YTD US 2Y yield increase (%ppt)



Source: J.P. Morgan

Figure 9: Data from the Tokyo exchange also points to a much cleaner position vs JPY relative to 2024...

Net open interest on FX daily futures contracts in the Tokyo Financial Exchange; net is long minus short contracts, scaled by contract unit size and converted to USDmn at market exchange rates.



Source: Bloomberg Finance L.P., Tokyo Financial Exchange, J.P. Morgan [Latam Local Markets Research](#)

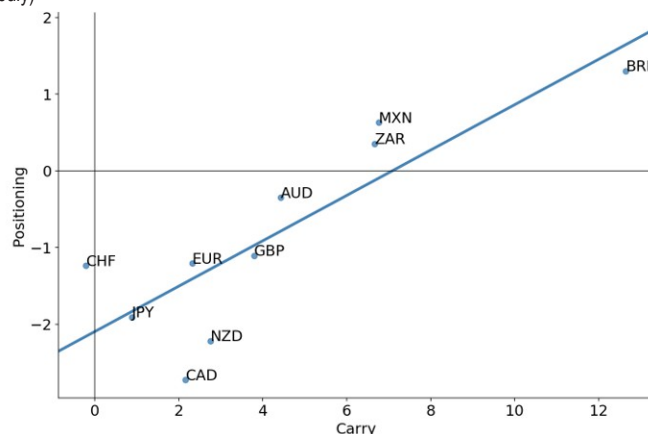
- **Positioning is still one-way.** Positioning is not a perfectly clean analog between 2024 and 2026 episodes, but on net we see more similarities than differences. It's true that some indicators suggest the yen-short trade isn't as entrenched as 2024 (Figure 9), but other metrics like CFTC indicate a similar-sized yen short. Meanwhile,

futures also indicate a concentrated carry position pre-intervention/FOMC (Figure 10). As such, **any additional shocks could force a further carry-positioning liquidation**; USD also screened long into both episodes.

- **Intervention is occurring against a generally pro-risk backdrop:** 2024 and 2026 carry performance evolved on the back of positive equity returns, with somewhat similar contours from 2Q-3Q, suggesting potentially similar risk appetite anchors for the carry trade.

Figure 10: ...though carry is still a relatively concentrated position in some corners of the markets

(X): Carry % (Y): Net FX futures positioning (3Y z-score). Pre-intervention (as of 28 July)



Source: J.P. Morgan, CFTC, Bloomberg Finance L.P.

Differences vs 2024

- **2026 doesn't have the same vol shock from a surprise BoJ hike.** Most remember MoF intervention after US CPI (discussed above). But importantly, the BoJ surprised market consensus by hiking not long thereafter, which accelerated the yen carry unwind and amplified the spillover into the cross-asset space (equities) reflecting the surprise nature. By contrast, BoJ chose not to hike last week (within 24hrs of the first intervention). It is also probably worth reminding that the surprise hike came much earlier in the BoJ normalization cycle, and alongside its surprise nature, probably fueled a greater overall financial tightening and deleveraging. **Overall this should help in limiting the collateral damage in 1) G10 FX, 2) the carry trade more broadly and 3) cross-asset vol like that of 2024.**
- **MoF's 2026 FX intervention has already far exceeded 2024 levels, which keeps risks elevated for 'new' sources intervention.** This has the market contemplating more intervention or intervention-like proxies, such as GPIF, and of course has prompted unprecedented US Treasury defense of the yen as well. While past MoF interventions

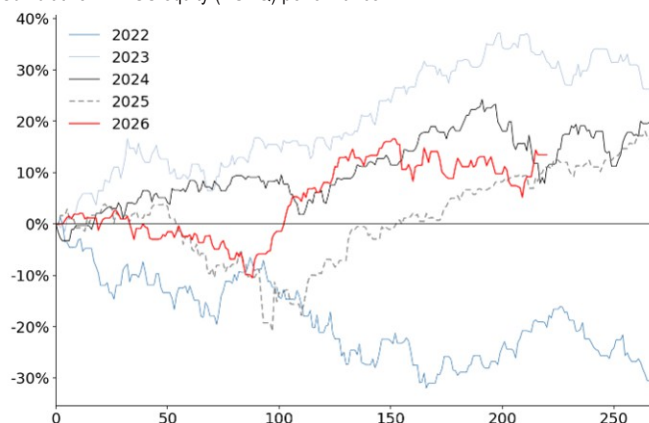
07 August 2026

proved fleeting, **questions linger about the duration and additional sources in the current episode.**

- **Politics are relevant in 2026, but not like the shock of 2024.** MXN was a consensus long in 2024, with exceptional carry (policy rate 11% mid-2024, vs 6.5% now). But there was a domestically-induced positioning shock from Claudia Sheinbaum's sweeping win with a congressional supermajority, that paved the way for potential constitutional changes. Thus, the long-side of the carry trade also came under particular pressure. **While BRL politics have jolted FX from time to time this year ahead of October elections, risks from elections through to the FX complex are likely more contained this year.**
- **Global growth backdrop on somewhat better footing in 2026.** Manufacturing PMIs continue to point to a strong global economy overall, with the global PMI tracking above 52 since April. Global PMIs were also mostly above 50 in 2024, but didn't breach 51, and also collapsed in July just as US labor markets were starting to soften. Thus, carry resilience could be attributed in part to stronger global cyclical footing this time around. **This points to more sustained carry resilience in this episode.**
- **ToT shocks are still evolving, and (mostly) a support.** 2026 of course will be remembered for the second great commodity vol shock of this cycle. Realized Brent vol in 2024 was 26%; this year is tracking 66%. We learned that the carry trade could withstand commodity shocks in 2022, anchored by widening dispersion of hiking cycles, but 2026 is also steadying carry returns by virtue of exporters tending to be the higher-yielding on average than lower-yielding peers (eg Asia, Europe). **Still, particularly sharp commodity vol spikes (eg early March) have forced some deleveraging, and so while we think the ongoing tension remains a carry support, flare-ups are still an intermittent tactical risk to the trade.**

Figure 11: Equity returns have tracked a similar contour through 2Q vs 2024

Cumulative YTD US equity (NSDQ) performance



Source: J.P. Morgan, Bloomberg Finance L.P.

Next week: RBA, Norges Bank, Riksbank; US CPI; Europe PMIs

Next 2 weeks will see **G10 central bank meetings** by the **RBA** (11th, JPM: on hold), **Norges Bank** (13th, JPM: on hold) and **Riksbank** (20th, JPM: on hold). **EM central bank meetings** will take place in **Romania** (10th, JPM: on hold), **Peru** (14th, JPM: on hold) and **Indonesia** (19th, JPM: on hold). Key inflation data will be released in **Norway** (10th), **US** (12th), **Canada** (17th), **UK** (19th), and **Japan** (21st). Labour market data to watch includes **Sweden** (13th), **UK** (18th) and **Australia** (20th). GDP data in **UK** (13th), **Euro area** (14th) and **Japan** (17th), along with PMIs in **Japan**, **Euro area** and **UK** (21st) will shed light on local growth narratives.

07 August 2026

Main trade recommendations

Trade type	Currency	Trade	P/L
Macro portfolio			
Cash (new trades)	—	—	-
Cash (existing)	CAD/MXN	— Sell CAD/MXN	3.60%
	AUD/CHF, USD/CHF, CHF/HUF	— Short CHF vs AUD, USD, HUF	0.27%
Options (new)	—	—	-
Options (existing)	NOK/SEK	— Buy NOK/SEK 3m call spread. K = 1.01/1.03. Spot ref. 1.0004. Cost 52bps.	0.00%
Closed	EUR/SEK	— Buy EUR/SEK	1.00%
	AUD/SEK	— Buy AUD/SEK	0.75%
	NZD/USD, USD/CAD, USD/SEK, USD/CHF, EUR/USD	— Buy USD vs basket of DM low yielders (NZD, CAD, SEK, CHF, EUR)	-0.69%
Derivatives portfolio			
RV (new trades)	USD/MXN	— Buy delta-hedged 3M USD/MXN ATM/25d 1*2 vega-neutral ratio call spreads	-0.4 vol pts.
RV (existing)	USD/ZAR	— Buy 2M in 4M USD/ZAR FVA	-1.3 vol pts.
	EUR/HUF-EUR/SEK	— Buy 3M 2% OTMS EUR puts/HUF calls vs. sell 3M 2% EUR puts/SEK calls	-19.7 bps.
	NZD/USD-EUR/NZD	— Buy 3M NZD/USD vol, financed by selling EUR/NZD vol	-0.3 vol pts.
	USD/CAD	— Buy 3M USD/CAD 1.45 digi call	-9.8%
RV (closed)	EUR/MXN	— Buy delta-hedged 3M EUR/MXN ATM/25d 1*2 vega-neutral ratio call spreads	2.7 vol pts.

Disclosures

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmm.com/markets> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan’s implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmm.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC (“JPMS”) acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation’s Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future

become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Notification for Credit Ratings: If this material includes credit ratings, such credit ratings provided by Japan Credit Rating Agency, Ltd. (JCR) and Rating and Investment Information, Inc. (R&I), are credit ratings provided by Registered Credit Rating Agencies (credit rating agencies registered under the Financial Instruments and Exchange Law of Japan (FIEL)). With respect to credit ratings that are provided by credit rating agencies other than JCR and R&I and have no stipulation that such credit ratings are provided by Registered Credit Rating Agencies, this means that such credit ratings are Non Registered Ratings (credit ratings provided by credit rating agencies not registered under the FIEL). Among the Non Registered Ratings, with respect to those credit ratings provided by S&P Global Ratings (S&P), Moody's Investors Service (Moody's), or Fitch Ratings (Fitch), prior to making investment decision based on such Non Registered Ratings, please carefully read the "Explanation Letter regarding Non Registered Ratings" for the corresponding credit rating agency, which we separately have sent or will send.

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material

has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpimipl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpimipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpimipl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan**

07 August 2026

Listed Securities”), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities. According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the “Important Disclosures” in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc (“JPMS plc”) which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited (“JPMML Ltd”) which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 (“the FPO”); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as “UK relevant persons”. This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA’s policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC (“JPMS”) is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst’s involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan’s current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts’ views and research, and it may also engage in transactions for its own account or for its clients’ accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this

transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein (“Information”) is reproduced by permission of MSCI Inc., its affiliates and information providers (“MSCI”) ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party (“J.P. Morgan Data”) in any third-party artificial intelligence (“AI”) systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 07 Aug 2026 12:29 PM EDT

Disseminated 07 Aug 2026 01:23 PM EDT